Problem Chosen	2024 HiMCM Summary Sheet	Team Control Number
B		15502

Summary

Against the backdrop of energy shortages and growing environmental concerns, the energy consumption and carbon emissions of HPC have become issues that attract widespread attention. In response, we conducted a study on the energy consumption and carbon footprint of HPC.

To assess the energy consumption of HPC, we designed an HPC energy consumption model that indirectly estimates energy use based on societal computing demands, overcoming the challenge of obtaining HPC energy consumption data. Our model first allocates the demand to two categories of systems: Massive Parallel Processing and cluster systems. Then, it proportionally distributes the computing tasks executed by these systems to individual high-performance computers and determines the global annual energy consumption of HPC based on their energy efficiency metrics. Through our computational analyses, we discern that the annual energy consumption of HPC under full load is approximately bounded within the range of $[1.97 \times 10^8, 2.87 \times 10^8]$ MWh, with the average annual consumption falling between $[0.72 \times 10^8, 1.43 \times 10^8]$ MWh.

To calculate the carbon emissions of HPC, we first developed an HPC carbon emissions calculation model that incorporates the energy mix, drawing on traditional methods for calculating carbon emissions from energy sources. Secondly, we utilized grey prediction models, resistance differential equations, and ARIMA models to forecast future trends in HPC energy consumption, HPC utilization rates, and seven major energy sources. By combining these forecasting models with the carbon emissions calculation model, we can determine the annual carbon emissions of HPC for the coming years. Our computational findings indicate that the projected carbon emissions for the year 2030 are poised to lie within the interval of $[8.18 \times 10^8, 10.54 \times 10^8]$ tons.

Expanding upon these models, we constructed an HPC carbon emission calculation model with the proportion of clean energy and nuclear energy as variables. By iterating through the increase of clean energy and nuclear energy proportions from 0 to 100%, we found that increasing the proportion of clean energy can effectively reduce carbon emissions. For instance, when the nuclear energy proportion is 20% and the new energy proportion is 60%, carbon emissions can be reduced by 77%.

Additionally, to study the impact of the increasing energy consumption of HPC on community energy rights, we took a distribution network that includes an HPC center as an example and used KCL and KVL to describe the relationship between the upper limit of community energy access and HPC energy consumption. By combining this with the aforementioned model's predictions for HPC energy consumption, we determined the upper limits of community energy access from 2024 to 2030. The results indicate that by 2028, the upper limit of community energy access is expected to drop below 80% of the 2024 level.

Finally, based on our research findings, we propose two recommendations: 1) Offer electricity price discounts to attract HPC companies to locate their servers in the ocean; 2) Provide power generation subsidies to encourage new energy enterprises to develop ocean-based renewable energy sources. By leveraging the abundant offshore new energy, we can offset the significant energy consumption of high-performance computing, which is beneficial for energy saving and carbon emission reduction.

Keywords: HPC; Energy Consumption; Carbon Emissions; Carbon Emissions Model; Forecasting Model; Island New Energy

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1 Introduction

1.1 Background

High Performance Computing (HPC) has become ubiquitous in today's society, with its formidable computing power driving revolutionary progress in multiple fields. For instance, in the field of artificial intelligence, HPC has made it possible to train large language and painting models; in the field of cloud computing, HPC has spurred the rapid development of big data analysis and online services[1]. Despite the advancements in society driven by HPC, it has also brought about numerous environmental issues, such as excessive energy consumption, generation of a large amount of electronic waste, constant noise production, and overexploitation of rare earth resources.

Amidst energy shortages and environmental degradation, the performance of HPC in terms of energy consumption and carbon emissions has garnered unprecedented attention. HPC facilities, especially large-scale data centers and supercomputers, are becoming new giants in energy consumption. To meet the growing computational demands, they not only exacerbate the pressure on energy supply but also have a significant impact on the environment. To ensure that HPC technology can serve the needs of future society in a more sustainable manner, it is essential to delve into its impact on energy consumption and carbon emissions and to develop innovative solutions.

1.2 Problem Restatement and Section Organization

Task	Problem Restatement	Organization
1	Assess the global annual energy consumption of high-performance computing from the perspectives of maximum and average loads	Section 2
2	Establish an HPC carbon emission model considering factors such as energy production methods and energy mix	Section 3
3	Explore the model changes when energy and HPC grow	
4	Predict the range of carbon emissions in 2030	
5	Discuss the reduction of HPC carbon emissions when the proportion of new energy increases	Section 4
6	Extend the model to another environmental factor	Section 5
7	Propose a set of recommendations to reduce the environmental impact of HPC	Section 6
8	Discuss how the proposed recommendation can be incorporated into the model	
9	Write a letter to the United Nations Consultative Committee	Letter

2 Part I: Assessing HPC Energy Consumption

2.1 Model Assumptions and Overview

When investigating the energy consumption of HPC, we found that there is currently a scarcity of data on HPC energy consumption. However, we have obtained information on the computing power required by society and the Top500 supercomputers[2], which provides detailed descriptions of the computing capabilities and energy consumption of typical HPC systems. Considering the

close relationship between energy consumption and the computing power demanded by society, we estimate the energy consumption of HPC indirectly through the computing power. To this end, we have made the following assumptions:

1	**Assumption**	The number of each computer in the Top500 list is the same worldwide.
	Justification	The exact number of each type of high-performance computer is difficult to obtain, but for modeling purposes, it is reasonable to assume a uniform distribution.
2	**Assumption**	The required computational power of society is borne by the 500 types of HPC.
	Justification	Given that the Top500 list covers the most powerful systems currently, the impact of other high-performance computers with lower performance on the overall computational demand is negligible.
3	**Assumption**	Each computer bears tasks proportional to its rated computing capacity.
	Justification	Although it is impossible to know the specific tasks allocated to each HPC category, it is reasonable to assume that they are allocated according to their capacity.

The modeling approach is illustrated in Figure 1. Firstly, we categorize these 500 models of HPC into two groups: Massive Parallel Processing (MPP) and Cluster systems. Secondly, we allocate the total computing power to these two broad categories. Then, we further distribute the computing power undertaken by MPP and Cluster systems to each individual HPC. Finally, by referring to the computing capabilities and energy conversion efficiency provided by the Top500 list, we calculate the energy consumption of each HPC.

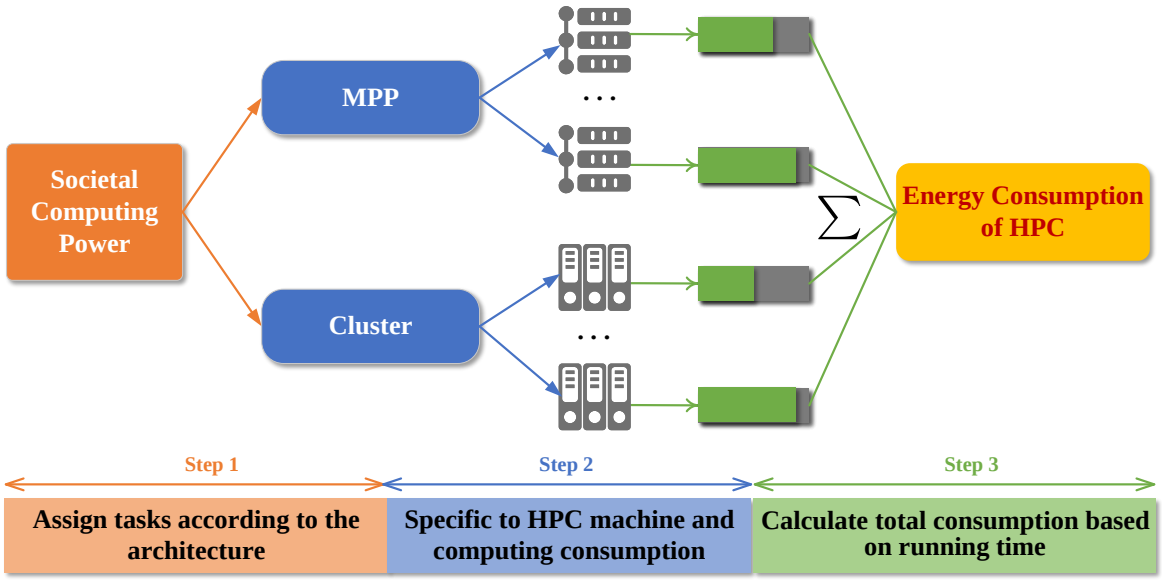


Figure 1: The framework diagramof HPC energy consumption model considering the computing power demand

2.2 HPC Energy Consumption Model Considering Computational Demand

We allocate the total computing demand of society evenly between MPP and Cluster based on their rated computing capabilities. The computing power that MPP and Cluster should undertake is shown in Equation 1.

$$\begin{aligned} T^b_{\text{MPP}} &= T_{\text{r}} \times \frac{\sum_{i=1}^{N_{\text{MPP}}} T^{\max}_{\text{MPP},i}}{\sum_{i=1}^{N_{\text{MPP}}} T^{\max}_{\text{MPP},i} + \sum_{i=1}^{N_{\text{Cluster}}} T^{\max}_{\text{Cluster},i}} \\ T^b_{\text{Cluster}} &= T_{\text{r}} \times \frac{\sum_{i=1}^{N_{\text{Cluster}}} T^{\max}_{\text{Cluster},i}}{\sum_{i=1}^{N_{\text{MPP}}} T^{\max}_{\text{Cluster},i} + \sum_{i=1}^{N_{\text{Cluster}}} T^{\max}_{\text{Cluster},i}} \end{aligned} \tag{1}$$

where T_{r} represents the computing power required by society, $T^{\max}_{\text{MPP},i}$ denotes the rated computing capacity of the i-th MPP-type HPC, N_{MPP} is the number of MPP-type HPCs, N_{Cluster} is the number of Cluster-type HPCs, T^b_{MPP} is the computing load that all MPP-type HPCs should bear, and T^b_{Cluster} is the computing load for all Cluster-type HPCs.

We also need to further allocate the computing power undertaken by MPP and Cluster to each individual HPC. Subsequently, we calculate the power of each HPC by weighting it with the energy efficiency parameters of each HPC, as shown in Equation 2.

$$\begin{aligned} P_{\text{MPP},i} &= T^b_{\text{MPP}} \times \frac{T^{\max}_{\text{MPP},i}}{\sum_{i=1}^{N_{\text{MPP}}} T^{\max}_{\text{MPP},i}} \times \frac{1}{E_{\text{MPP},i}} \\ P_{\text{Cluster},i} &= T^b_{\text{Cluster}} \times \frac{T^{\max}_{\text{Cluster},i}}{\sum_{i=1}^{N_{\text{MPP}}} T^{\max}_{\text{Cluster},i}} \times \frac{1}{E_{\text{Cluster},i}} \end{aligned} \tag{2}$$

where $E_{\text{MPP},i}$ represents the energy efficiency of the i-th MPP-type HPC, $E_{\text{Cluster},i}$ denotes the energy efficiency of the i-th Cluster-type HPC, $P_{\text{MPP},i}$ is the power of the i-th MPP-type HPC, and $P_{\text{Cluster},i}$ is the power of the i-th Cluster-type HPC.

When calculating the annual energy consumption of global HPC at full load, simply multiply power by time, with the unit of energy consumption power being megawatts (MW). However, we also note that in addition to the computational energy consumption of the HPC itself, the normal operation of the HPC also requires auxiliary systems such as cooling and power distribution, and the energy consumption of these systems should also be included. The annual energy consumption of the HPC is shown in Equation 3.

$$\begin{aligned} ECS_{\text{Full}} &= (ECS^{\text{MAP}}_{\text{Full}} + ECS^{\text{Cluster}}_{\text{Full}}) \times PUE \\ &= (\sum_{i=1}^{N_{\text{MPP}}} P_{\text{MPP},i} \times 8760 + \sum_{i=1}^{N_{\text{MPP}}} P_{\text{Cluster},i} \times 8760) \times PUE \end{aligned} \tag{3}$$

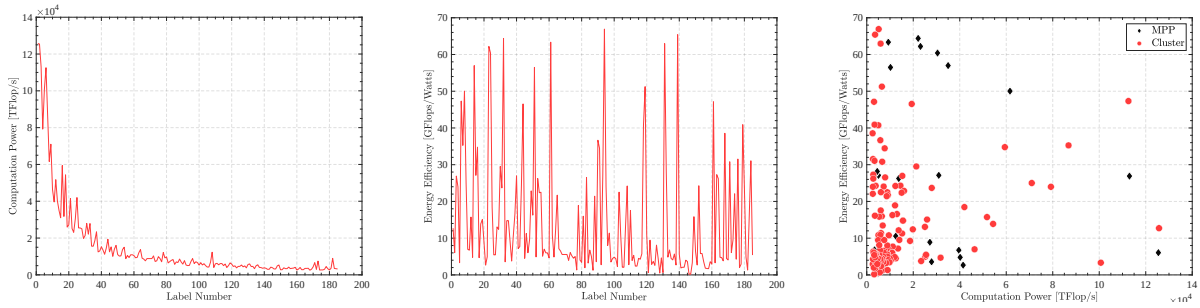
where PUE stands for Power Usage Effectiveness, 8760 represents the total number of hours in a year, and ECS_{Full} denotes the annual energy consumption of the HPC.

Additionally, we note that global HPCs do not operate at full capacity throughout the year[3]. Therefore, it is necessary to incorporate the average utilization rate AUR to calculate the average annual energy consumption ECS_{Average} of global HPCs.

$$ECS_{\text{Average}} = (\sum_{i=1}^{N_{\text{MPP}}} P_{\text{MPP},i} \times 8760 + \sum_{i=1}^{N_{\text{MPP}}} P_{\text{Cluster},i} \times 8760) \times AUR \times PUE \tag{4}$$

2.2.1 Model Solution and Result Analysis

Firstly, we collected the TOP500 dataset and performed data cleaning to eliminate some missing data, ultimately obtaining relevant data for 187 high-performance computers, including computing power, energy efficiency, architectural type, etc., as shown in Figures 2a, 2b, and 2c. After consulting various sources, we found that by the end of 2023, the total computing power of society was approximately 551 EFlops[4], and the range of values for PUE is [1,2] [5]. Newly built or retrofitted HPC clusters can achieve below 1.5. Incorporating these results into Equation 2 allows us to determine the specific computing power that MPP and Cluster should undertake. Subsequently, by applying these values into Equations 3 and 4, we can calculate the maximum annual energy consumption and the average annual energy consumption of HPCs.



(a) Computing Power (b) Energy Efficiency (c) Architecture Type

Figure 2: Data related to the Top 500

The annual energy consumption of HPC is related to the PUE indicator, average utilization rate, and the total computing power of society. Therefore, we discussed the annual energy consumption of HPC when PUE varies within the range of $[1, 2]$ and the total computing power of society varies within $[300, 900]$ EFlops, with the results shown in Table 2 and Figure 2. Additionally, we discussed the annual energy consumption of HPC when the average utilization rate varies within the range of $[0.3, 0.9]$ and the total computing power of society varies within $[300, 900]$ EFlops, with the results shown in Table 1 and Figure 3.

In Table 2, the increase from top to bottom represents the average utilization rate rising from 0.3 to 0.8, and the increase from left to right represents the societal computing power increasing from 300 EFlops to 900 EFlops. This indicates that higher average utilization rates and larger scales of societal computing power lead to greater annual energy consumption of HPCs. In Table 1, the higher the PUE value, the more additional energy is required for cooling, power distribution,

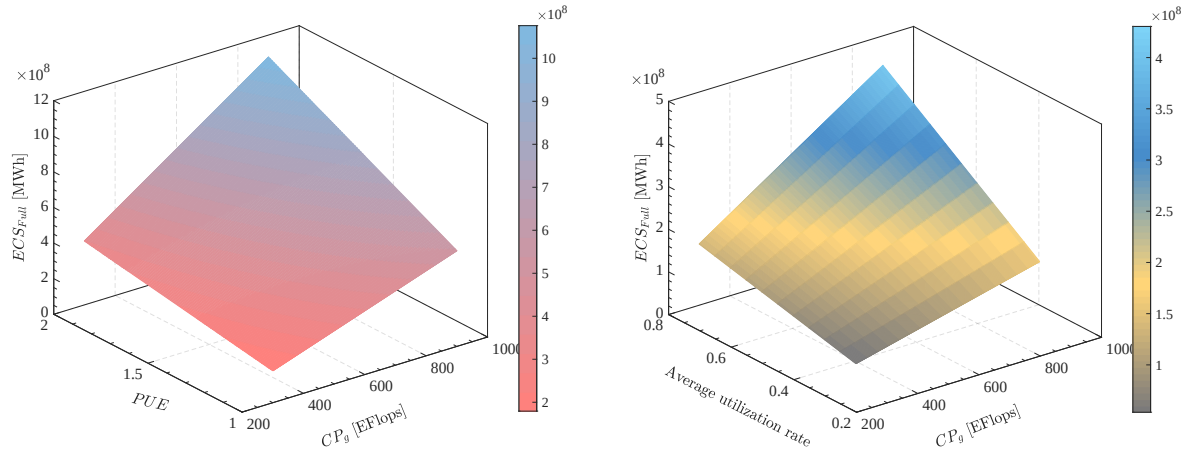
Table 1: Global Annual Energy Consumption of HPC under Full Load

PUE	Required Computing Power [EFlops]						
	300	400	500	600	700	800	900
1	1.79×10^8	2.39×10^8	2.99×10^8	3.59×10^8	4.18×10^8	4.78×10^8	5.38×10^8
1.166	2.09×10^8	2.79×10^8	3.49×10^8	4.18×10^8	4.88×10^8	5.58×10^8	6.28×10^8
1.332	2.39×10^8	3.19×10^8	3.98×10^8	4.78×10^8	5.58×10^8	6.38×10^8	7.17×10^8
1.498	2.69×10^8	3.59×10^8	4.48×10^8	5.38×10^8	6.28×10^8	7.17×10^8	8.07×10^8
1.664	2.99×10^8	3.98×10^8	4.98×10^8	5.98×10^8	6.97×10^8	7.97×10^8	8.96×10^8
1.83	3.29×10^8	4.38×10^8	5.48×10^8	6.57×10^8	7.67×10^8	8.77×10^8	9.86×10^8

Table 2: Global Annual Energy Consumption of HPC Considering Average Utilization Rate

AUR	Required Computing Power [EFlops]						
	300	400	500	600	700	800	900
0.3	5.38×10^7	6.28×10^7	7.17×10^7	8.07×10^7	8.97×10^7	9.86×10^7	1.08×10^8
0.4	7.17×10^7	8.37×10^7	9.56×10^7	1.08×10^8	1.20×10^8	1.32×10^8	1.43×10^8
0.5	8.97×10^7	1.05×10^8	1.20×10^8	1.34×10^8	1.49×10^8	1.64×10^8	1.79×10^8
0.6	1.08×10^8	1.26×10^8	1.43×10^8	1.61×10^8	1.79×10^8	1.97×10^8	2.15×10^8
0.7	1.26×10^8	1.46×10^8	1.67×10^8	1.88×10^8	2.09×10^8	2.30×10^8	2.51×10^8
0.8	1.43×10^8	1.67×10^8	1.91×10^8	2.15×10^8	2.39×10^8	2.63×10^8	2.87×10^8

and other systems to maintain normal operation, demonstrating that more energy is consumed in scenarios with lower energy efficiency.



(a) Full Load Energy Consumption (b) Average Energy Consumption

Figure 3: Full Load and Average Energy Consumption of HPC

From Figure 3a, it can be observed that as the demand for societal computing power and the PUE increase, the full load energy consumption tends to rise. A lower PUE value indicates lower energy consumption for the same computing demand, signifying higher energy efficiency of HPCs. Figure 3b reveals that with the increase in computing demand and utilization rate, the average energy consumption also increases. In recent years, the average utilization rate of HPCs has contin-

ued to rise slowly with the expansion of the total societal computing power, indicating that there is still room for growth in the total demand for computing power.

Considering the current range of PUE values from 1.1 to 1.6, the annual energy consumption under full load for HPCs can be calculated to be approximately within $[1.97 \times 10^8, 2.87 \times 10^8]$ MWh. Referring to the average utilization rate range of [0.35, 0.5], the annual average consumption of HPCs can be calculated to be approximately within $[0.69 \times 10^8, 1.44 \times 10^8]$ MWh.

3 Part II: Assessing HPC Carbon Emissions

3.1 Model Assumptions and Overview

To assess the carbon emissions of HPC by 2030, we have made the following assumptions and developed two models: 1) the HPC carbon emission model considering the energy mix, and 2) the computing power and energy forecast model. The HPC carbon emission model uses HPC energy demand and energy mix as inputs and outputs the total carbon emissions. The computing power and energy forecast model is responsible for predicting the computing power and various energy situations that HPC will undertake in the coming years. By integrating these two models, we can obtain the annual carbon emissions of HPC for the subsequent years.The modeling approach is illustrated in Figure 4.

1	**Assumption**	Only seven types of energy sources are considered: coal, gas, oil, wind, solar, hydropower, and nuclear energy.
	Justification	According to the energy data published in 2024, these seven energy sources account for the largest proportion, and other energy sources can be neglected.
2	**Assumption**	The carbon emissions generated from the production of clean energy equipment, such as wind turbines and photovoltaic panels, are averaged over total produced electricity of their lifecycle.
	Justification	The latest research indicates that although wind turbines and photovoltaic panels do not produce carbon dioxide during operation, their production does generate carbon dioxide, which should be distributed over their lifecycle.
3	**Assumption**	The electricity demand of HPC grows exponentially.
	Justification	In recent years, HPC has developed rapidly, especially after the emergence of ChatGPT in 2022, showing an overall exponential growth pattern.

3.2 Submodel 1: HPC Carbon Emission Calculation Model Considering Energy Mix

We have observed that by 2023, fossil fuels such as coal, oil, and natural gas account for approximately 60% of the total electricity generation, while hydropower, nuclear energy, and renewable sources like wind and solar account for about 39.2% [6]. These seven sources essentially constitute the global electricity generation. Therefore, we disregard other forms of energy and primarily consider these seven types. The total carbon emissions caused by HPC in year t are shown in Equation 5.

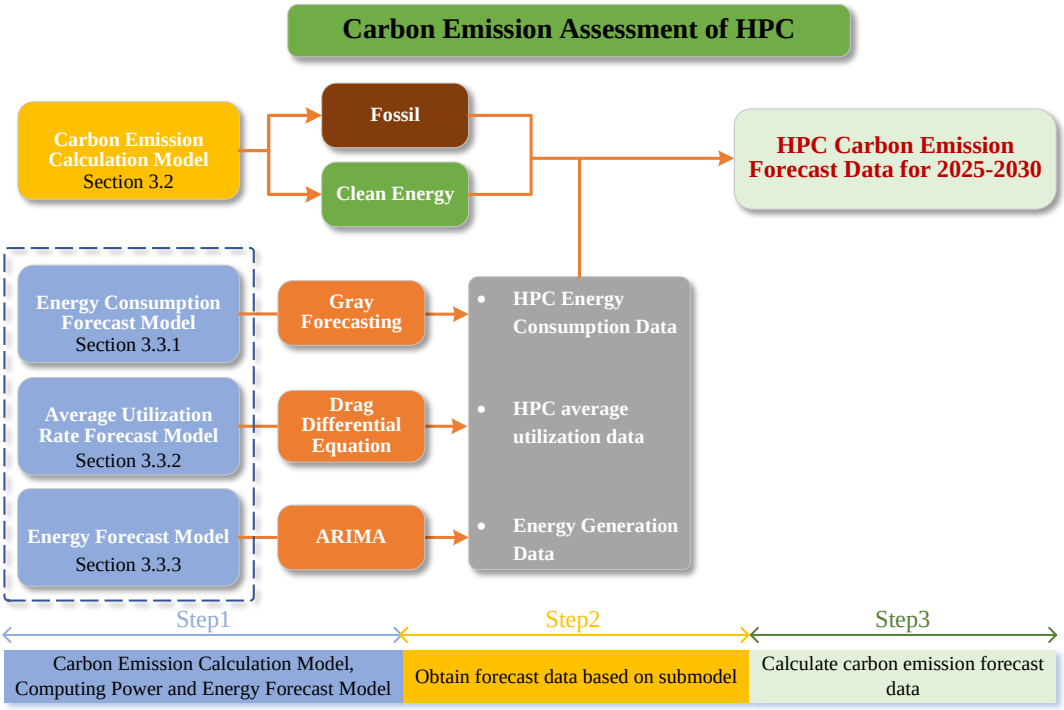


Figure 4: The framework diagram of HPC Carbon Emission Model

$$\begin{aligned} CE_j = & f_{\text{Coal}}\left(G_t^{\text{Coal}}\right) + f_{\text{Oil}}\left(G_t^{\text{Oil}}\right) + f_{\text{Gas}}\left(G_t^{\text{Gas}}\right) \\ & + f_{\text{Wind}}\left(G_t^{\text{Wind}}\right) + f_{\text{Solar}}\left(G_t^{\text{Solar}}\right) + f_{\text{Hydro}}\left(G_t^{\text{Hydro}}\right) + f_{\text{Nuclear}}\left(G_t^{\text{Nuclear}}\right) \end{aligned} \tag{5}$$

where G_t^{Coal}, G_t^{Oil}, G_t^{Gas}, G_t^{Wind}, G_t^{Solar}, G_t^{Hydro}, and G_t^{Nuclear} represent the electricity generation from Coal, Oil, Gas, Wind, Solar, Hydro, and Nuclear in year t, respectively. The functions $f_{\text{Coal}}(\cdot)$, $f_{\text{Oil}}(\cdot)$, $f_{\text{Gas}}(\cdot)$, $f_{\text{Wind}}(\cdot)$, $f_{\text{Solar}}(\cdot)$, $f_{\text{Hydro}}(\cdot)$, and $f_{\text{Nuclear}}(\cdot)$ represent the relationship between the electricity generation and carbon emissions of the aforementioned energy sources. We will provide a detailed explanation of these functions next.

- **Carbon Emissions from Fossil Fuels:** Coal, Oil, Gas

  The carbon emissions from fossil fuel power generation can be roughly calculated using emission factors. The process of fossil fuel power generation essentially involves converting thermal energy into mechanical energy by power generation units, which is then used for electricity generation. The efficiency of the units varies under different operating conditions, affecting the CO2 emissions per unit of electricity generated [7]. Therefore, the carbon emissions from fossil fuel power generation cannot be simply multiplied by the emission factor; they must also be adjusted based on the operating conditions. The formula for the

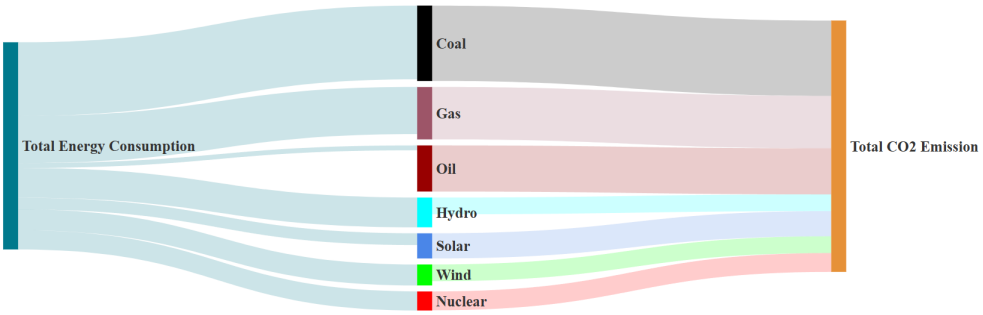


Figure 5: HPC Carbon Emission Calculation Approach

carbon emissions from fossil fuels is shown in Equation 6.

$$\begin{aligned} f_{\text{Coal}}\left(G_t^{\text{Coal}}\right) &= \eta \cdot k_{\text{Coal}} \cdot G_t^{\text{Coal}} \\ f_{\text{Coal}}\left(G_t^{\text{Oil}}\right) &= \eta \cdot k_{\text{Oil}} \cdot G_t^{\text{Oil}} \\ f_{\text{Gas}}\left(G_t^{\text{Gas}}\right) &= \eta \cdot k_{\text{Gas}} \cdot G_t^{\text{Gas}} \end{aligned} \tag{6}$$

$$\eta = a \times P^3 + b \times P^2 + c \times P + d \tag{7}$$

In the equations, $k_{\text{Coal}} = 820$ g/kWh is the carbon emission coefficient for coal power, $k_{\text{Oil}} = 720$ g/kWh is the carbon emission coefficient for oil, and $k_{\text{Gas}} = 490$ g/kWh is the carbon emission coefficient for gas [8, 9]. The correction factor η is a polynomial curve related to the load P of the power generation unit [10], as shown in Figure 6.

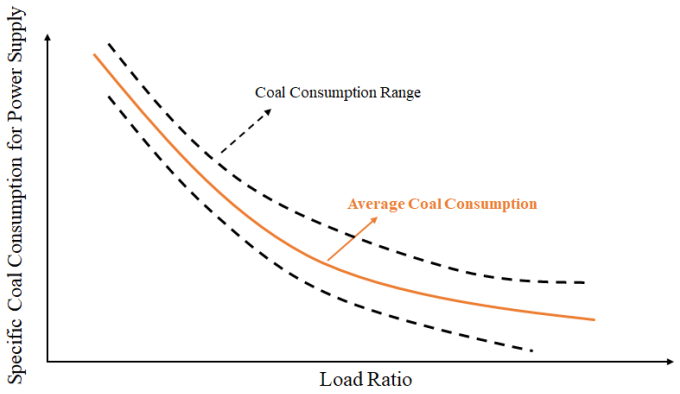


Figure 6: Correction factors for carbon emission coefficients of fossil energy

- **Carbon Emissions from Clean Energy:** Wind, Solar, Hydro

  New energy sources such as wind, solar, and hydro power do not produce carbon dioxide during electricity generation and are considered completely clean. However, some studies indicate that considering new energy power generation as having zero carbon emissions is not reasonable. It is necessary to account for the carbon emissions associated with the entire

lifecycle of new energy equipment, including production and transportation [11]. Therefore, when calculating the carbon emissions from new energy power generation, we adopt the approach of allocating the full lifecycle carbon emissions to the expected electricity generation within the design service life of the equipment, calculated as the levelized carbon dioxide emissions, as shown in Equation 8.

$$\begin{aligned} f_{\text{Wind}}\left(G_t^{\text{Wind}}\right) &= \frac{CE_{\text{m}}^{\text{Wind}}}{EP_{\text{use}}^{\text{Wind}}} \times G_t^{\text{Wind}} \\ f_{\text{Solar}}\left(G_t^{\text{Solar}}\right) &= \frac{CE_{\text{m}}^{\text{Solar}}}{EP_{\text{use}}^{\text{Solar}}} \times G_t^{\text{Solar}} \\ f_{\text{Solar}}\left(G_t^{\text{Hydro}}\right) &= \frac{CE_{\text{m}}^{\text{Hydro}}}{EP_{\text{use}}^{\text{Hydro}}} \times G_t^{\text{Hydro}} \end{aligned} \tag{8}$$

where $CE_{\text{m}}^{\text{Wind}}$, $CE_{\text{m}}^{\text{Solar}}$, and $CE_{\text{m}}^{\text{Hydro}}$ represent the carbon emissions associated with the production and transportation of the equipment over their entire lifecycle. $EP_{\text{use}}^{\text{Wind}}$, $EP_{\text{use}}^{\text{Solar}}$, and $EP_{\text{use}}^{\text{Hydro}}$ represent the total electricity generated by the equipment over their service life.

- **Carbon Emissions from Nuclear Energy**

  Nuclear energy harnesses the heat generated from nuclear fission of fuel in a reactor, which is then transferred to a steam generator by a coolant, heating the coolant to a high temperature and pressure state. The high-temperature, high-pressure steam drives the rotation of a turbine, which in turn drives the generator to produce electricity, without emitting carbon dioxide into the atmosphere. However, it is also necessary to consider the carbon emissions over its entire lifecycle, as shown in Equation 9.

$$f_{\text{Nuclear}}\left(G_t^{\text{Nuclear}}\right) = \frac{CE_{\text{m}}^{\text{Nuclear}}}{EP_{\text{use}}^{\text{Nuclear}}} \times G_t^{\text{Nuclear}} \tag{9}$$

  where $CE_{\text{m}}^{\text{Nuclear}}$ represents the carbon emissions associated with the production and transportation of nuclear power plants over their entire lifecycle, and $EP_{\text{use}}^{\text{Nuclear}}$ represents the total electricity generated by the equipment over its service life.

3.3 Submodel 2: Computing Power and Energy Forecast Model

3.3.1 HPC Full Load Energy Consumption Forecast Model

HPC is a cutting-edge technology with limited available data, but it is developing rapidly, showing an exponential growth trend. In response to the characteristics of scarce data and rapid development, we have decided to use the grey prediction model to forecast the full load energy consumption of HPC. The GM(1,1) grey prediction model is a method used to establish an approximate model for sequences that basically meet the conditions for forming differential equations, in order to predict subsequent data [12]. Assuming the original sequence is $w^{(0)}(k) = (w^{(0)}(1), w^{(0)}(2), \cdots, w^{(0)}(n))$, the original sequence is accumulated to obtain the cumulative generating sequence, and a differential equation is constructed for the cumulative generating sequence, as follows:

$$\begin{cases} w^{(1)} = \left\{ w^{(1)}(1), w^{(1)}(2), w^{(1)}(3), ...w^{(1)}(n) \right\} \\ \frac{\mathrm{d}w^{(1)}}{\mathrm{d}t} + \alpha w^{(1)} = \mu \end{cases} \tag{10}$$

where α represents the development grey number, and μ represents the internal control grey number.

Next, we solve Equation 10. Let $Y_n = [w^{(0)}(2), w^{(0)}(3), \cdots, w^{(0)}(n)]^T$, and the parameters are obtained using the least squares method.

$$\widehat{\partial} = (B^T B)^{-1} B^T Y n \tag{11}$$

where $\widehat{\partial}$ represents the vector of parameters to be estimated, and B represents the intermediate quantity; their expressions are as follows:

$$B = \begin{pmatrix} -\frac{1}{2}(w^{(1)}(1) + w^{(1)}(2)) & 1 \\ -\frac{1}{2}(w^{(1)}(2) + w^{(1)}(3)) & 1 \\ \vdots & \vdots \\ -\frac{1}{2}(w^{(1)}(n-1) + w^{(1)}(n)) & 1 \end{pmatrix}, \widehat{\partial} = \begin{pmatrix} a \\ u \end{pmatrix}$$

Solving Equation 11 yields the discrete-time response function of the grey prediction, and then restoring the predicted values of the accumulated sequence allows us to obtain the prediction for the original sequence, as shown in Equation 12.

$$\begin{cases} w^{(1)}(t+1) = [w^{(0)}(1) - \frac{u}{a}]e^{-at} + \frac{u}{a} \\ \widehat{w}^{(0)}(t+1) = \widehat{w}^{(1)}(t+1) - \widehat{w}^{(1)}(t) \end{cases}, t = 0, 1, 2 \cdots n-1 \tag{12}$$

where $\widehat{w}(t+1)$ represents the predicted value of the accumulated sequence.

3.3.2 HPC Average Utilization Rate Forecast Model

With the popularization and development of technology, the average utilization rate of HPC will show an increasing trend, but the growth rate will gradually approach zero. This is because the total amount of electricity available in society is finite, and other industries also require energy, competing with HPC for resources, eventually reaching a state of equilibrium. In other words, the average utilization rate of HPC will increase gradually in a decelerating manner, eventually approaching a fixed value. These characteristics are very similar to the SI model used in the study of infectious diseases, hence we establish a forecast model for the average utilization rate of HPC based on the SI model [13].

Assume that the average utilization rate of HPC in year t is $AUR(t)$, and the growth rate of the average utilization rate in year t is $\lambda(t)$. Then, within the time interval $[t, t+\Delta t]$, the increase in the average utilization rate of HPC is shown in Equation 13.

$$AUR(t+\Delta t) - AUR(t) = \lambda(t) AUR(t) \Delta t \tag{13}$$

Considering the competitive relationship for energy resources among various industries, the growth rate of the average utilization rate of HPC, $\lambda(t)$, cannot be a fixed constant, as this would lead to an unlimited increase in the average utilization rate of HPC. Therefore, we assume that $\lambda(t)$ decreases gradually and is linearly related to the decrease in $AUR(t)$, as follows:

$$\lambda(t) = \lambda_0 (0.8 - AUR(t)) \tag{14}$$

where λ_0 is the initial growth rate.

By dividing both sides of Equation 13 by Δt and taking the limit, we obtain the differential equation for predicting the average utilization rate of HPC, as shown in Equation 15.

$$\frac{dAUR(t)}{dt} = \lambda_0 (0.8 - AUR(t)) AUR(t) \tag{15}$$

3.3.3 Energy Forecast Model

We employ the ARIMA model to forecast the electricity generation of various energy sources in the coming years. This is because 1) electricity generation often exhibits significant seasonality and trends, such as wind and solar power being influenced by seasonal and weather conditions; 2) electricity generation data are often non-stationary, meaning their statistical properties (like mean and variance) change over time. The ARIMA model combines three methods: Autoregression (AR), Integration (I), and Moving Average (MA), making it particularly suitable for processing and forecasting non-stationary time series data. Below is a detailed explanation of the mathematical principles of the ARIMA model [14]:

- **Autoregression (AR):** The autoregressive part focuses on the linear relationship between the current value and past values in a time series. If the current value Y_t of a time series can be expressed as a linear combination of its past p values plus a random error term ϵ_t, then the time series has autoregressive properties.

  $$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p} + \epsilon_t \tag{16}$$

  where $\phi_1, \phi_2, \ldots, \phi_p$ are the autoregressive coefficients.

- **Integration (I):** The integration part is the process of transforming a non-stationary time series into a stationary one. The number of differences d indicates how many times differencing is needed to make the series stationary. First differencing $\Delta Y_t = Y_t - Y_{t-1}$, second differencing $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$, and so on.

- **Moving Average (MA):** The moving average part focuses on the linear relationship between the current value and past error terms in a time series. If the current value Y_t of a time series can be expressed as a linear combination of the current and past q error terms, then the time series has moving average properties. The mathematical expression is:

  $$Y_t = \mu + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-q} + \epsilon_t \tag{17}$$

  where μ is the mean of the series, and $\theta_1, \theta_2, \ldots, \theta_q$ are the moving average coefficients.

The ARIMA model integrates the above three components, and its general form can be expressed as:

$$\phi(B)(1-B)^d Y_t = \theta(B)\epsilon_t \tag{18}$$

where $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_p B^p$ is the autoregressive polynomial, $\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \ldots + \theta_q B^q$ is the moving average polynomial, and B is the backshift operator, such that $BY_t = Y_{t-1}$.

3.4 Model Solution and Result Analysis

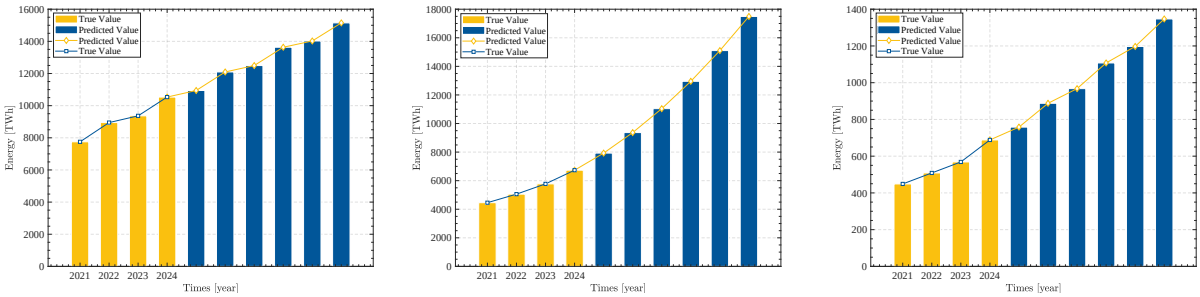
Firstly, we collected the electricity generation data of fossil fuels and new energy sources from 2021 to 2024 [15]. We input this dataset into the ARIMA model as shown in Equation 18 to predict the generation trends of various energy sources up to 2030. The results are presented in Table 3 and Figures 7 and 8, which allow us to forecast the changes in the global electricity generation energy mix for the coming years. Concurrently, we obtained the scale data of HPC for the past four years and substituted this data into the grey prediction model as Equation 12 to obtain the HPC computing power scale from 2021 to 2030 [16]. Together with the predicted energy mix data, these were input into the proposed HPC energy consumption model to derive the annual full load energy consumption of HPC for the coming years, as shown in Figure 9a. On the other hand, based on the average utilization rate data of HPC and the utilization rate forecast model, we were able to calculate the predicted data as shown in Figure 9b. By substituting into Equation 4, we could calculate the predicted annual average energy consumption data.

Table 3: Changes and forecasts of fossil energy from 2021 to 2030

	Power generation [TWh]						
	2024	2025	2026	2027	2028	2029	2030
Coal	10533.60	10942.56	12095.74	12494.74	13634.06	14023.11	15148.62
Gas	6733.13	7929.90	9365.64	11040.111	12953.09	15104.37	17493.72
Oil	688.28	758.12	887.82	967.67	1107.37	1197.22	1346.92
Wind	4488.75	4798.00	5705.75	6025.00	6942.75	7272.00	8199.75
Solar	1496.25	1634.96	1911.70	2039.54	2294.71	2411.75	2646.22
Hydro	4249.35	4547.36	4803.96	5017.75	5190.96	5327.06	5429.81
Nuclear	2723.18	2852.87	3042.42	3182.12	3381.67	3531.37	3740.92

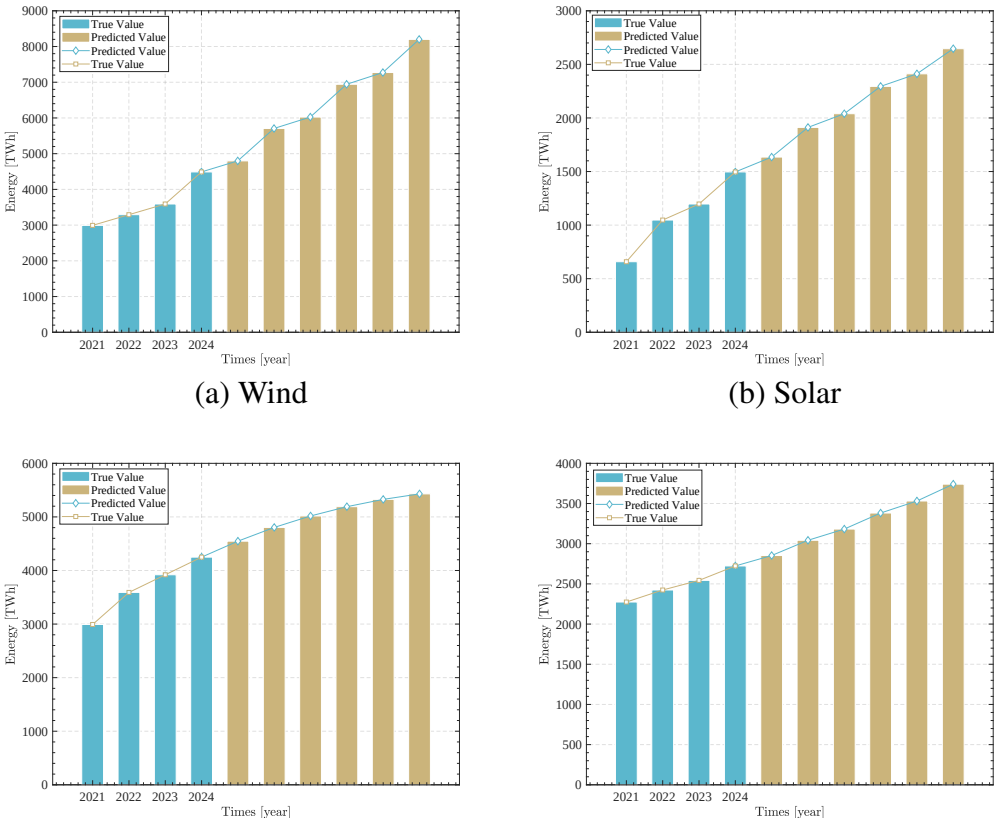
From Table 3 and Figure 8, it can be observed that the electricity generation from coal and oil continues to rise. However, when considering the proportion within the energy mix, their share of electricity generation is showing a declining trend. Meanwhile, the generation from natural gas is increasing significantly and is expected to surpass coal as the primary fuel for electricity generation by 2029. Renewable energy sources such as wind and solar power are developing rapidly and will capture a larger share of the global electricity generation mix in the coming years. Hydropower, being a mature resource for electricity generation, exhibits a slower growth trend in generation, nearing its maximum potential. Nuclear energy, as an emerging technology for power generation with high associated costs, still holds a small proportion of the total electricity generation but is showing a slow and steady growth pattern.

From Figure 9a, it can be observed that the development trend of HPC is positive, with the projected energy consumption in 2030 expected to reach five times that of 2024. Concurrently, as



(a) Coal (b) Gas (c) Oil

Figure 7: Changes and forecasts of fossil fuels from 2021 to 2030



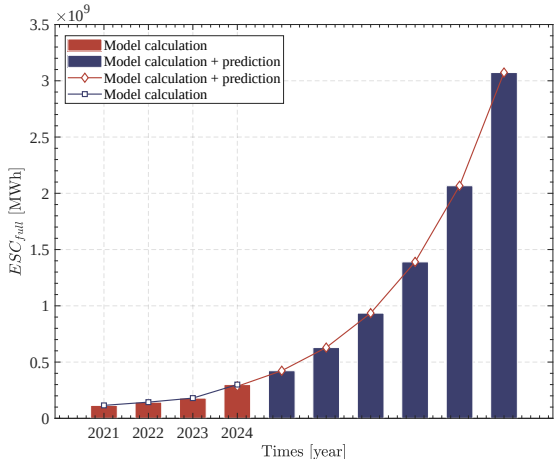
(a) Wind (b) Solar

(c) Hydro (d) Nuclear

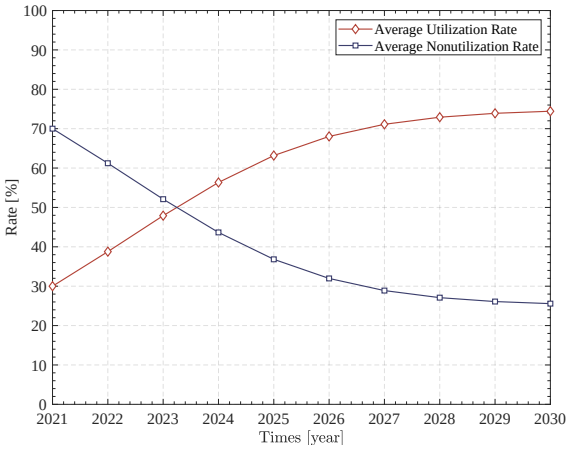
Figure 8: Changes and forecasts of clean energy from 2021 to 2030

the application of HPC expands on a large scale and the technology matures, Figure 9b shows that the average utilization rate will gradually increase, reaching a critical peak of approximately 75%.

In Tables 4 and Figure 11, we present the range of carbon emission forecasts for the years 2024-2030. It can be observed that as the demand for societal computing power grows, energy consumption increases rapidly. However, the green transition in the global electricity generation energy mix is not yet mature, leading to a continued upward trend in carbon emissions. By 2030,



(a) Annual Full Load Energy Consumption of HPC



(b) Average Utilization Rate

Figure 9: Variations and Forecasts of HPC Energy Consumption and Utilization Rate from 2021 to 2030

Table 4: Carbon Emission Forecast for 2024-2030

	Carbon emissions [ton]						
	2024	2025	2026	2027	2028	2029	2030
Maximum	7.3×10^7	1.2×10^8	1.9×10^8	3.0×10^8	4.6×10^8	7.0×10^8	10.5×10^8
Average	6.5×10^7	1.1×10^8	1.7×10^8	2.7×10^8	4.1×10^8	6.2×10^8	9.4×10^8
Minimum	5.7×10^7	9.5×10^7	1.5×10^8	2.4×10^8	3.6×10^8	5.4×10^8	8.2×10^8

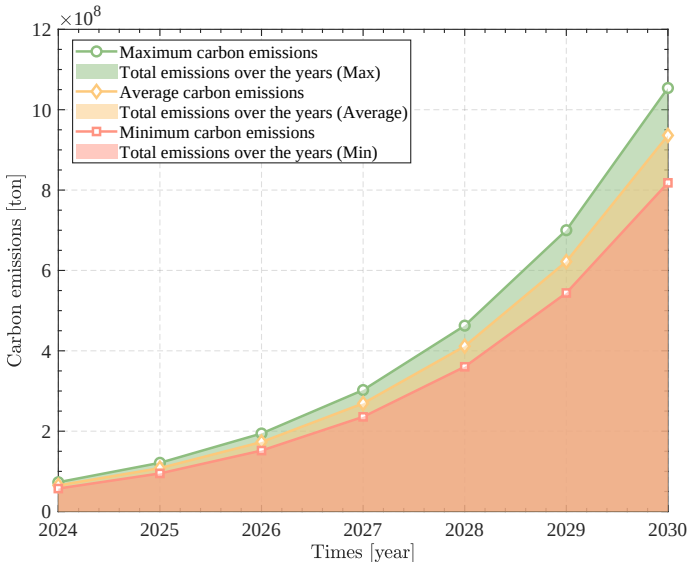


Figure 10: Carbon Emission Forecast for 2024-2030

the emissions are projected to exceed the current levels by more than tenfold, approximately rang-

ing from 8.18×10^8 to 10.54×10^8 tons of carbon dioxide equivalent.

4 Part III: Discussing the Impact of Increasing the Share of New Energy

We have categorized the seven types of energy discussed above into three categories based on their nature: 1) Fossil fuels, including coal, oil, and natural gas; 2) New energy sources, including wind, solar, and hydropower; 3) Nuclear energy. In this section, we discuss the impact of the proportion of these three types of energy on carbon emissions based on the calculation results for 2030 from the models above.

Let the electricity generation from coal, natural gas, and oil account for the fossil fuel electricity generation as ρ^1_{fossil}, ρ^2_{fossil}, and ρ^3_{fossil}, respectively. Then the average carbon emissions from fossil fuel power generation are shown in Equation 19.

$$CE^{\text{average}}_{\text{fossil}} = \frac{f_{\text{Coal}}\left(G^{\text{Coal}}_{2030}\right)}{G^{\text{Wind}}_{2030}} \cdot \rho^1_{\text{fossil}} + \frac{f_{\text{Gas}}\left(G^{\text{Gas}}_{2030}\right)}{G^{\text{Gas}}_{2030}} \cdot \rho^2_{\text{fossil}} + \frac{f_{\text{Oil}}\left(G^{\text{Oil}}_{2030}\right)}{G^{\text{Oil}}_{2030}} \cdot \rho^3_{\text{fossil}} \tag{19}$$

where the specific values of ρ^1_{fossil}, ρ^2_{fossil}, and ρ^3_{fossil} are based on the United Nations' forecast for various energy sources in 2030.

Let the electricity generation from wind, solar, and hydropower account for the clean energy generation as ρ^1_{clean}, ρ^2_{clean}, and ρ^3_{clean}, respectively. Then the average carbon emissions from clean energy power generation are shown in Equation 20.

$$CE^{\text{average}}_{\text{clean}} = \frac{f_{\text{Wind}}\left(G^{\text{Wind}}_{2030}\right)}{G^{\text{Wind}}_{2030}} \cdot \rho^1_{\text{clean}} + \frac{f_{\text{Solar}}\left(G^{\text{Solar}}_{2030}\right)}{G^{\text{Solar}}_{2030}} \cdot \rho^2_{\text{clean}} + \frac{f_{\text{Hydro}}\left(G^{\text{Hydro}}_{2030}\right)}{G^{\text{Oil}}_{2030}} \cdot \rho^3_{\text{clean}} \tag{20}$$

Let ρ_{fossil} be the proportion of fossil fuels, ρ_{clean} be the proportion of clean energy, and ρ_{nuclear} be the proportion of nuclear energy, then the carbon emissions of HPC are shown in Equation 21.

$$CE_{2030} = (CE^{\text{average}}_{\text{fossil}} \cdot \rho_{\text{fossil}} + CE^{\text{average}}_{\text{clean}} \cdot \rho_{\text{clean}}) \cdot ESC^{2030}_{\text{Average}} + f_{\text{Nuclear}}\left(\rho_{\text{nuclear}} \cdot ESC^{2030}_{\text{Average}}\right) \tag{21}$$

where $ESC^{2030}_{\text{Average}}$ is the average annual energy consumption of HPC in 2030, calculated from the above models, as shown in Figure 9.

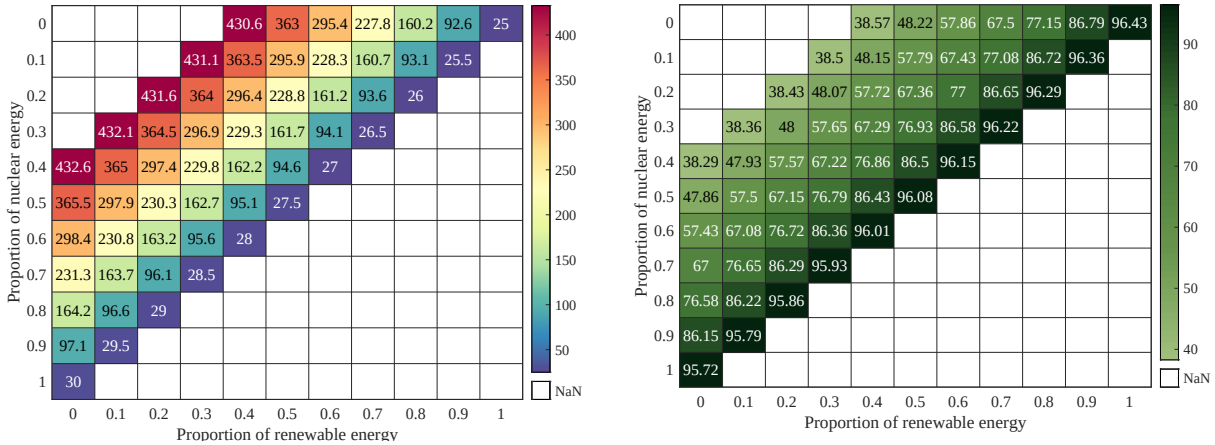
Since $\rho_{\text{fossil}} + \rho_{\text{clean}} + \rho_{\text{nuclear}} = 1$, Equation 21 can be rewritten as follows:

$$\begin{aligned} CE_{2030} = {} & (CE^{\text{average}}_{\text{fossil}} \cdot (1 - \rho_{\text{clean}} - \rho_{\text{nuclear}}) + CE^{\text{average}}_{\text{clean}} \cdot \rho_{\text{clean}}) \cdot ESC^{2030}_{\text{Average}} \\ & + f_{\text{Nuclear}}\left(\rho_{\text{nuclear}} \cdot ESC^{2030}_{\text{Average}}\right) \end{aligned} \tag{22}$$

Taking the carbon emissions of HPC for 2030 from the above model as the baseline value $\widehat{CE}_{2030}$, we can obtain the carbon emission reduction after changing the proportion of new energy and nuclear energy, as shown in Equation 23.

$$CE_{\text{reduce}} = \frac{\widehat{CE}_{2030} - CE_{2030}}{\widehat{CE}_{2030}} \times 100 \tag{23}$$

We discuss the reduction in carbon emissions and the percentage reduction when ρ_{clean} varies within $[0, 1]$ and ρ_{nuclear} varies within $[0, 1]$, as shown in Figure 11. The horizontal axis of the graph represents the proportion of new energy, the vertical axis represents the proportion of nuclear energy, the numbers in the grid represent the carbon emissions and the percentage reduction compared to the baseline scenario, and grids without numbers indicate no reduction in carbon emissions.



(a) Carbon Emissions under Different Proportions

(b) Percentage reduction in carbon emissions

Figure 11: Reduction in Carbon Emissions with the Growth of New Energy and Nuclear Energy

From Figure 11, it can be seen that as the proportion of renewable energy and nuclear power generation increases, carbon emissions significantly decrease, and the increase in the proportion of renewable energy generation brings more significant emission reduction benefits. When the power generation structure fully shifts to renewable energy and nuclear power, carbon emissions will drop to 4%-5% of the current level, essentially achieving green computing for HPC.

After our research, we found that using 100% new energy to power HPC would bring the following potential issues:

1. **Unstable Power Supply:** Meteorological factors such as wind speed and solar radiation are random and uncertain, which determines that new energy cannot provide a stable power supply. For example, when there is no wind or illumination at night, new energy equipments cannot generate power, which is fatal for HPC and may lead to server shutdowns.

2. **Voltage Instability:** Traditional thermal power units are considered voltage sources and can provide stable support for grid voltage, while new energy often adopts grid-following control and is considered a current source, unable to support grid voltage. Therefore, when

the proportion of new energy is high, the grid's ability to resist disturbances weakens, which may lead to potential power outage incidents.

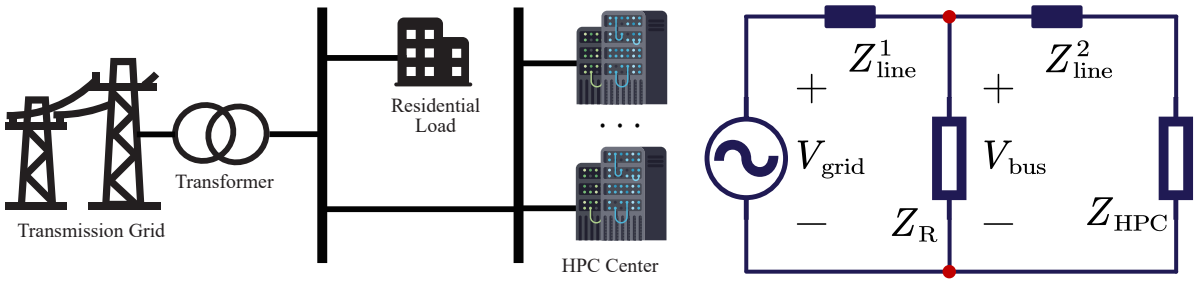
3. **Massive Energy Storage Required:** To overcome the problems brought by the randomness of new energy, energy storage is often needed. Currently, the cost of energy storage is high and its lifespan is short. When the proportion of new energy is high, a large amount of energy storage is required, which not only creates significant costs but also faces maintenance and recycling issues for the storage devices in the next decade.

5 Part V: Extending the Model to Other Key Areas

5.1 Model of HPC's Impact on Community Energy Rights

We have chosen to investigate the upper limit of energy access for local communities affected by HPC facilities. The energy demand of HPC facilities may exert pressure on local power grids, leading to unstable electricity supply and even power shortages. This tension in energy distribution may force electricity providers to prioritize the needs of HPC facilities over those of residential and small commercial users, thereby affecting employment and the healthy development of the community economy. We consider the upper limit of energy that can be accessed as a right, and this section addresses the impact of HPC's high energy consumption on the energy rights of communities.

A simple power system containing an HPC center is shown in Figure 13, where the transmission grid transmits electrical energy to the distribution grid through transformers, and communities and HPC centers are connected to the distribution grid busbar to obtain electrical energy. Figure 13b is the equivalent circuit diagram, where V_{grid} represents the output voltage of the transformer, V_{bus} is the distribution grid busbar voltage, Z^1_{line} and Z^2_{line} represent line impedances, Z_{R} represents the equivalent impedance of residential loads, and Z_{HPC} represents the equivalent impedance of the HPC center. Due to the voltage division effect of Z^1_{line}, V_{bus} will be lower than V_{grid}, but standards stipulate that V_{bus} should not be lower than $0.9V_{\text{grid}}$, which determines that the power obtained by communities and HPC centers is capped.



(a) System Diagram (b) Equivalent Circuit

Figure 12: Power system including HPC center

First, we derived the currents in each branch, and then, according to Kirchhoff's current law (KCL), which states that the sum of currents entering a node equals the sum of currents leaving

the node, we constructed a system of algebraic equations. After simplification, we established the relationship between V_{bus} and Z_{HPC}, as shown in Equation 24.

$$\begin{cases} I^1_{\text{line}} = I^2_{\text{line}} + I_{\text{R}} \\ I^1_{\text{line}} = \frac{V_{\text{grid}} - V_{\text{bus}}}{Z^1_{\text{line}}} \\ I^2_{\text{line}} = \frac{V_{\text{bus}}}{Z^2_{\text{line}} + Z_{\text{R}}} \\ I_{\text{R}} = \frac{V_{\text{bus}}}{Z_{\text{HPC}}} \end{cases} \quad \Longrightarrow \quad \frac{V_{\text{grid}} - V_{\text{bus}}}{Z^1_{\text{line}}} = \frac{V_{\text{bus}}}{Z_{\text{R}}} + \frac{V_{\text{bus}}}{Z^2_{\text{line}} + Z_{\text{HPC}}} \tag{24}$$

Setting $V_{\text{bus}} = 0.9V_{\text{grid}}$, we can calculate the relationship between Z_{R} and Z_{HPC} under maximum transmission conditions, as shown in Equation 25.

$$Z_{\text{R}} = \frac{9Z^1_{\text{line}}\left(Z^2_{\text{line}} + Z_{\text{HPC}}\right)}{Z^2_{\text{line}} + Z_{\text{HPC}} - 9Z^1_{\text{line}}} \tag{25}$$

Equation 25 uses an impedance model to equivalently represent the loads of residential and HPC center loads, which is not intuitive. Considering $P = \frac{V^2}{Z}$, Equation 25 is rewritten in terms of power variables, as shown in Equation 26.

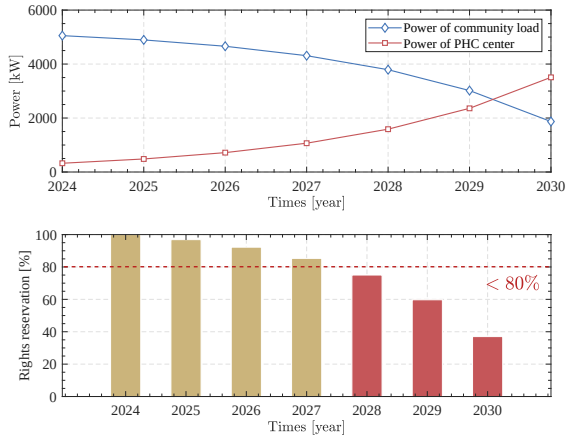
$$P_{\text{R}} = \frac{Z^2_{\text{line}} + \frac{\left(V^{ref}_{\text{bus}}\right)^2}{P_{\text{HPC}}} - 9Z^1_{\text{line}}}{9Z^1_{\text{line}}\left(Z^2_{\text{line}} + \frac{\left(V^{ref}_{\text{bus}}\right)^2}{P_{\text{HPC}}}\right)} \times \left(V^{ref}_{\text{bus}}\right)^2 \tag{26}$$

where V^{ref}_{bus} is the rated value of the busbar voltage.

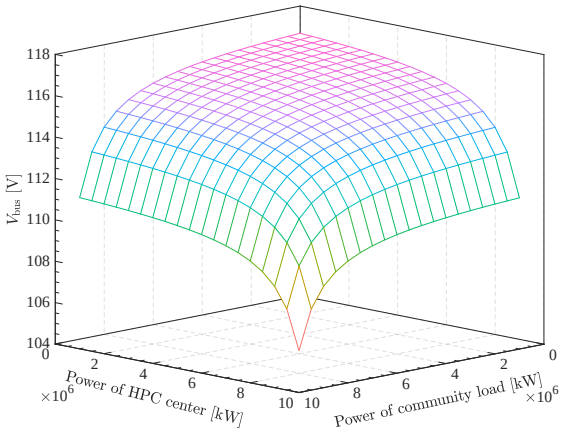
5.2 Model Solution and Result Analysis

We have constructed a regional distribution network and discussed the changes in HPC center load from 2024 to 2030, as well as the changes in residential and commercial power under voltage stability constraints, as shown in Figure 13a, where the vertical axis of the bar chart represents the percentage of residential and commercial power compared to 2024. At the same time, we analyzed the fluctuation of busbar voltage when HPC center load and residential commercial power load change, as shown in Figure 13b.

From Figure 13a, it can be observed that without upgrading the power supply capacity of the regional distribution network, as the HPC center load increases year by year, the electricity available to community residents shows an accelerating downward trend. By 2028, their electricity rights are less than 80% of 2024, causing inconvenience to residential electricity use. By 2030, it drops to below 40%, severely affecting the production and life of residents and local businesses. Moreover, when both are at their peak, the busbar voltage will drop below 105V, far exceeding the 5% voltage drop standard set by the NEC, which may lead to a collapse of the local power grid and cause significant economic losses.



(a) HPC centre load - residential commercial power relationship



(b) Bus voltage-load relationship

Figure 13: Impact of HPC centre loads on power systems

6 Recommendations of Policy and Technology

6.1 Recommendations to Reduce the Environmental Impact of HPC

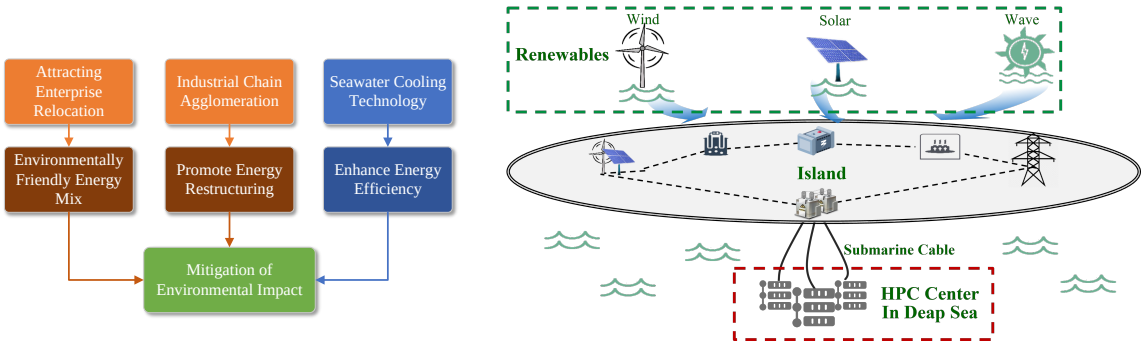
We have noticed that some top technology companies are already using island new energy to power their undersea HPC centers, and at the same time using seawater as a cooling medium, setting an excellent example for improving PUE and reducing the carbon emission intensity of HPC [17, 18]. Policies and technological developments can be leveraged to guide the layout of HPC towards islands/deep-sea development, and thus we propose the following three recommendations.

1. **Attracting Enterprise Relocation Policies:** For enterprises willing to relocate their HPC centers to islands or deep-sea areas where new energy sources are more abundant, offer a discount subsidy on electricity prices based on electricity consumption.

2. **Industrial Chain Agglomeration Policies:** For enterprises willing to build new energy power generation facilities for the HPC industry in areas with abundant new energy sources such as islands or deep-sea areas, purchase electricity at a higher price to help enterprises quickly recoup investments.

3. **Developing Seawater Cooling Technology:** Provide HPC centers with better and more cost-effective cooling systems, enhancing the energy efficiency of HPC centers.

The first two policy recommendations aim to use electricity price subsidies to attract the HPC industry to relocate to island/deep-sea areas rich in new energy resources, making the energy mix of the HPC industry more environmentally friendly. At the same time, they encourage new energy companies to accelerate the construction of local new energy networks, promote energy restructuring, and reduce carbon emissions from electricity use. The third suggestion, tailored to the unique environment of islands/deep-sea areas, involves the development of distinctive seawater



(a) Recommendation Diagram (b) HPC with Energy Island

Figure 14: Impact of Recommendations

cooling technology to enhance the energy efficiency of HPC centers and further reduce the energy consumption of HPC. The predicted effects are shown as figure 14b.

6.2 Model Upgrade

Assuming that the aforementioned Recommendation 1 is adopted, since the HPC centers are relocated to areas with a richer supply of new energy, their electricity energy structure will contain a greater proportion of renewable sources such as wind and solar power, corresponding to a reduction in carbon emissions. Therefore, it is necessary to re-model the power generation energy structure and forecasting models for island and other new energy-rich areas, replacing the global energy structure and forecasting models in the original models, as shown in the table below.

Table 5: Model Upgrade Comparison

Model	Origin	Upgraded
Section 2	Considering that the proposed policy has no impact on the model	
Section 3.2	Considering HPC power	**Consider net power of the island**
Section 3.3.1	Considering that the proposed policy has no impact on the model	
Section 3.3.2	Considering that the proposed policy has no impact on the model	
Section 3.3.3	Consider Global Energy Trend	**Consider Energy developing on Island Grid**
Section 5.1	Consider the structure	**Consider the island grid structure**

7 Strengths and Weaknesses

Strengths

1. **Innovative Estimation Method.** Our model indirectly estimates the energy consumption of HPC by the required computing power of society, innovatively linking computational demand with energy consumption, solving the problem of difficult access to direct energy consumption data of HPC.

2. **Comprehensiveness.** Our model takes into account a variety of energy types and calculates

carbon emissions based on the characteristics of each type of energy, enabling the model to more accurately assess the impact of different energy mixes on HPC carbon emissions.

3. **Adaptability.** Our model has predictive capabilities, able to forecast future changes in HPC energy consumption and energy mix, which helps to more precisely calculate the carbon footprint of HPC.

Weaknesses

1. **Dependence on Assumptions.** Our model makes many assumptions of uniform distribution, such as the number of each type of computer in the TOP500 list being consistent, and the computing power undertaken by each type of computer being evenly distributed according to rated computing power. These assumptions deviate from reality.

2. **Data Limitations.** Our model is limited by the quality and scope of available data. For instance, specific energy consumption data for HPC and lifecycle carbon emission data for new energy devices, if incomplete or outdated, can affect the model's predictive capabilities and the reliability of the results.

3. **Lack of Technical Details.** When assessing the impact of HPC energy consumption on community energy rights, we simplified the AC power grid into a DC grid because we only learned about DC models during high school, which certainly deviates from the actual AC power grid.

8 Conclusion

This study presents a comprehensive assessment model for the energy consumption and carbon emissions of High Performance Computing (HPC), aiming to provide a quantitative analysis of the environmental impact of HPC and to explore the potential effects of increasing the share of new energy on reducing carbon emissions. Here are the main conclusions from our research:

1. HPC Energy Consumption Assessment: We found that the annual global energy consumption of HPC is closely related to the Average Utilization Rate (AUR), Power Usage Effectiveness (PUE), and the total scale of societal computing demands. By analyzing scenarios with different PUE values and scales of computing demands, we concluded that the annual energy consumption of HPC under full load ranges roughly between $[1.97 \times 10^8, 2.87 \times 10^8]$ MWh, while the average annual consumption falls between $[0.69 \times 10^8, 1.44 \times 10^8]$ MWh.

2. HPC Carbon Emissions Model: Considering the energy mix, we predict that by 2030, the carbon emissions of HPC will exceed the current level by more than ten times, ranging from 8.18×10^8 to 10.54×10^8 tons of carbon dioxide equivalent. This forecast highlights the potential environmental impact of HPC, especially in the context of a global electricity structure that has not yet fully transitioned to green energy.

3. Impact of New Energy Proportion on Carbon Emissions: Our model indicates that as the proportion of new energy and nuclear power generation increases, the carbon emissions of

HPC significantly decrease. When the power generation structure fully shifts to new energy and nuclear power, carbon emissions will drop to 4%-5% of the current level, essentially achieving green computing for HPC.

4. Impact of HPC on Community Energy Rights: We constructed an electrical power system model to analyze the impact of HPC facilities on the upper limit of energy access for local communities. The results show that without upgrading the power supply capacity of the regional distribution network, as the load of HPC centers increases, the electricity available to community residents exhibits an accelerating downward trend, dropping below 40% of 2024 levels by 2030, severely affecting the production and life of residents and local businesses.

5. Policy and Technology Recommendations: Our three recommendations, Attracting Enterprise Relocation Policies, Industrial Chain Agglomeration Policies, and the development of Seawater Cooling Technology aim to use electricity price subsidies to attract the HPC industry to relocate to island/deep-sea areas rich in new energy resources, while encouraging new energy companies to accelerate the construction of local new energy networks, promote energy restructuring, and reduce carbon emissions from electricity use.

In conclusion, based on our research findings, we have written a letter to the United Nations Consultative Committee urging them to improve the section on the environmental impact of high-performance computing in their development goals for 2030.

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9 letter

Dear Members of the United Nations Advisory Board,

We are writing to you in light of the recent report titled Governing AI for Humanity, which was issued in September 2024. We commend the Advisory Board for its comprehensive approach to AI governance; however, we note with concern that the report does not significantly address the environmental impacts of High Performance Computing (HPC), a critical component of modern AI systems.

Our research reveals that HPC, while indispensable in advancing fields like AI, climate science, and data analytics, also poses substantial environmental challenges. For instance, the annual global energy consumption of HPC, when operating at full load, ranges between 1.97×10^8 and 2.87×10^8 MWh. This energy demand not only strains global power resources but also contributes heavily to carbon emissions, with projections suggesting that by 2030, HPC carbon emissions could exceed current levels by over tenfold, reaching up to 10.54×10^8 tons of carbon dioxide equivalents annually.

Moreover, our findings indicate that a gradual transition to cleaner energy sources within the HPC sector could substantially mitigate its environmental impact. If HPC centers shift to renewable energy sources like wind, solar, and nuclear power, emissions could be reduced by up to 95%. However, such a transition requires targeted policies to encourage HPC centers to relocate to areas rich in renewable resources and adopt energy-efficient cooling technologies like seawater cooling. Specific measures, such as providing electricity subsidies to HPC centers that integrate local renewable energy sources and expanding support for green technology development, could be pivotal in reducing the carbon footprint of HPC.

Beyond environmental concerns, the unchecked growth of HPC infrastructure may also strain community energy resources. In regions where HPC facilities are sited, our models predict a potential reduction in local community energy access to as low as 40% of the current levels by 2030, threatening the energy security and quality of life of local populations. These findings emphasize the urgency of including HPC in sustainable development goals and call for policies that support equitable access to energy resources.

To conclude, we respectfully ask the Advisory Board to recognize HPCs environmental impact within its 2030 goals. By addressing HPC in upcoming developmental agendas, the United Nations can lead the world toward responsible and sustainable technological advancement.

Thank you for considering our recommendations. We look forward to seeing a more inclusive approach to HPC and AI governance in future reports.

Sincerely yours,

Your friends